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EVEN SEMESTER EXAMINATION JUNE- 2025

Name of the Course: B.Tech
Name of the paper: Design of Machine Elements-II
Time: 3 hours

Semester: VI
Paper Code: TME 602
Maximum Marks: 100

Note:-

(i) All questions are compulsory
(ii) Answer any two sub questions among a, b, & c in each main question
(iii) Total marks in each main question are twenty
(iv) Each question carries 10 marks

Q1.	**20 marks**	
(a)	Derive the expression for deflection and maximum shear stress in a close-coiled helical spring under axial load.	**CO 1**
(b)	Discuss the effects of eccentric loading, buckling, and surge in helical springs. How do these phenomena influence the stress distribution, stability, and dynamic performance of the spring?	
(c)	It is required to design a helical torsion spring for a window shade. The spring is made of patented and cold-drawn steel wire of Grade-4. The yield strength of the material is 65% of the ultimate tensile strength, and the factor of safety is 2.5. Due to space constraints, the mean coil diameter is maintained at 22 mm. The maximum bending moment acting on the spring is 300 N-mm. The modulus of elasticity of the spring material is 210,000 N/mm^2, and the required stiffness of the spring is 4 N-mm/rad. Determine the following design parameters: (i) Wire diameter (d) (ii) Number of active coils (n) (iii) Maximum shear stress in the spring wire (iv) Total length of the spring wire required (v) Deflection of the spring under maximum load	
Q2.	**20 marks**	
(a)	Explain the different types of belts used in mechanical power transmission systems. Compare flat belts and V-belts in terms of construction, working principle, and applications.	**CO 1**
(b)	A belt drive transmits power from a pulley of 500 mm diameter rotating at 180 RPM. The coefficient of friction between the belt and the pulley is 0.30, and the angle of lap on the smaller pulley is 170°. The maximum allowable tension in the belt is 2200 N. Determine: (i) The velocity of the belt (ii) The ratio of belt tensions (iii) The effective tension in the belt (iv) The power transmitted by the belt	
(c)	Explain the working principle of a V-belt drive and derive the expression for the ratio of tensions considering the groove angle.	**CO 2**
Q3.	**20 marks**	
(a)	Derive Buckingham's equation for wear strength of gear teeth. Explain the significance of each term involved.	**CO 3**

(b)	A pair of spur gears with 20° full-depth involute teeth consists of a 20-teeth pinion meshing with a 41-teeth gear. The module is 3 mm, and the face width is 40 mm. The material for both the pinion and gear is steel with an ultimate tensile strength of 600 N/mm^2, and the gears are heat-treated to a surface hardness of 400 BHN. The pinion rotates at 1450 rpm, and the service factor for the application is 1.75. Assume that the velocity factor accounts for the dynamic load, and the factor of safety is 1.5. Determine the rated power that the gears can transmit.	
(c)	List and explain the different types of failure in gears. Discuss the causes, effects, and preventive measures for each type.	
Q4.	**20 marks**	
(a)	Explain and derive the force analysis of a helical gear. Show how the normal, tangential, radial, and axial components of the tooth force are determined.	**CO 3**
(b)	A pair of parallel helical gears consists of a 24-teeth pinion rotating at 5000 rpm and transmitting 2.5 kW of power to the gear. The speed reduction is 4:1. The normal pressure angle and helix angle are 20° and 23°, respectively. Both gears are made of hardened steel with an ultimate tensile strength (Sut) of 750 N/mm^2. The service factor is 1.5, and the factor of safety is 2. The gears are finished to meet the accuracy of Grade 4. (i) In the preliminary stage of design, assume the velocity factor accounts for the dynamic load and the face width is ten times the normal module. If the pitch line velocity is 10 m/s, estimate the normal module. (ii) Select the first preference value of the normal module (as per standard series) and calculate the main dimensions of the gears.	
(c)	Perform force analysis for worm gear drive and derive equations for tangential, radial, and axial forces acting on the worm and worm wheel.	**CO 4**
Q5.	**20 marks**	
(a)	For rolling contact bearings, derive the expression for static and dynamic load capacity. Explain how equivalent load is determined for combined radial and axial loads.	
(b)	A single-row deep groove ball bearing is subjected to a radial load of 8 kN and an axial (thrust) load of 3 kN. The shaft rotates at 1200 rpm, and the desired bearing life L_{10h} is 20,000 hours. The minimum acceptable shaft diameter is 75 mm. Select a suitable ball bearing from a manufacturer's catalogue that meets these requirements. Clearly show the steps involved in calculating the equivalent dynamic load, required dynamic load capacity, and bearing selection.	**CO 5**
(c)	Differentiate between hydrodynamic bearings and hydrostatic (static) bearings in terms of their working principles, lubrication mechanisms, applications, and advantages. Support your answer with neat sketches.	**CO 6**